Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Some previous assignments that will help me complete this project is the crypto corgi assignment. It will be useful to reference the drafter aspect of it to help navigate me through the project. I plan to heavily use conditionals and loops in this project where dataclasses will be useful in assigning different properties to a specific element. The programing skill that I want to improve on is route functions and drafter related code.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The part of the project that I expect to be the easiest for me is creating functions that will be used in the drafter functions. I expect the hardest part of the project will be programing the drafter section because I feel like I never fully understood how to program it.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I estimate the drafter section will require the most time and effort. I think dataclasses of the necessary elements should be implemented first and then functions involving bonding.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions that might interfere with my work is creating unnecessary functions when I can implement them into one. I will minimize this by using the LLM to help me simplify them. Resources that I will need available to me while working is previous assignments, a periodic table, and the LLM.

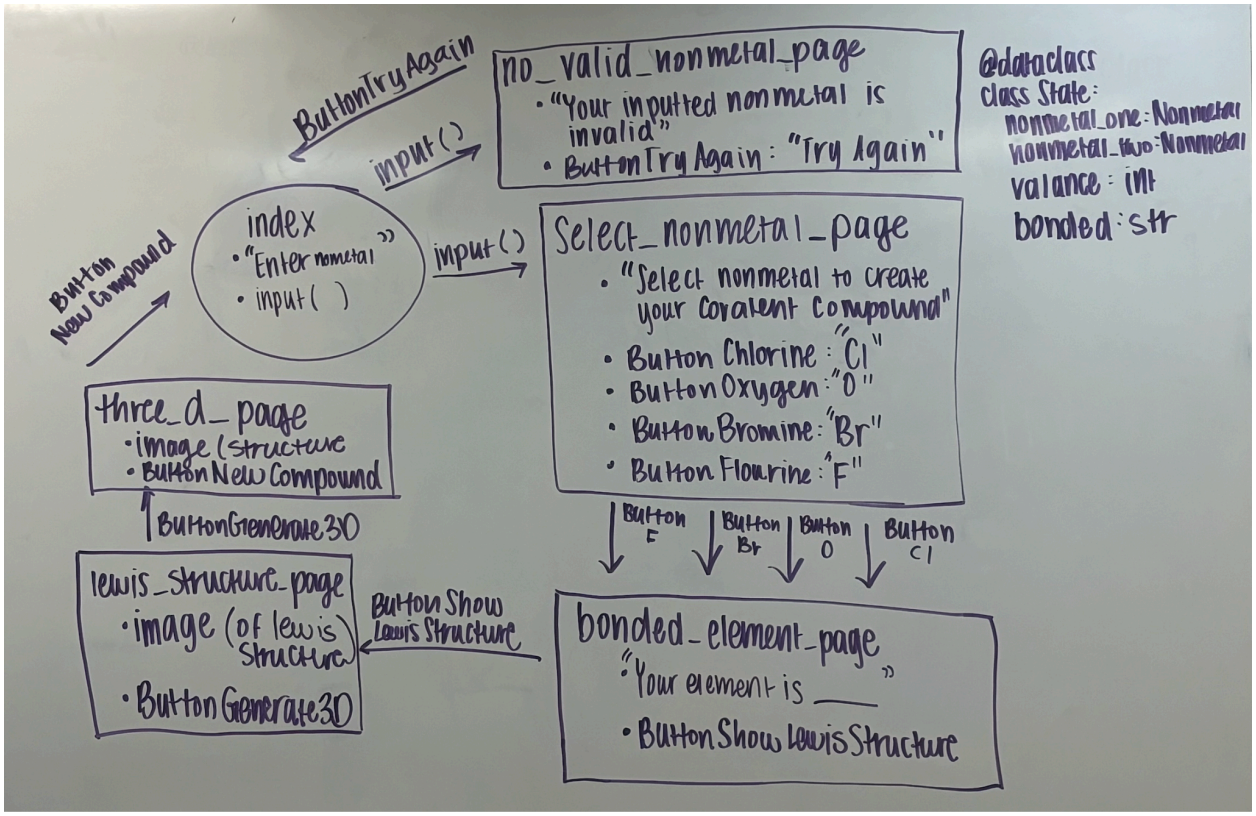
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


ButtonTryAgain
input()
no_valid_nonmetal_page
• "Your inputted nonmetal is invalid"
• ButtonTryAgain: "Try Again"
@dataclass
class State:
nonmetal_one: Nonmetal
nonmetal_two: Nonmetal
valance: int
bonded: str
Button New Compound
index
• "Enter nonmetal"
• input( )
input()
Select_nonmetal_page
• "Select nonmetal to create your Covalent Compound"
• Button Chlorine: "Cl"
• Button Oxygen: "O"
• Button Bromine: "Br"
• Button Flourine: "F"
three_d_page
• image (structure
• Button New Compound
ButtonGenerate3D
Button F
Button Br
Button O
Button Cl
lewis_structure_page
• image (of lewis structure)
• Button Generate3D
Button Show Lewis Structure
bonded_element_page
"Your element is ____"
• Button Show Lewis Structure

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of this site is to model how covalent compounds are made and create a real life application where it can model the lewis structure as well as the 3D model.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	**The 4 fields include:** ● **non_metal_one** ○ **This represents the first nonmetal being bonded. Its type will be a dataclass Nonmetal which will include its Symbol, Valance Electrons, and number of electrons to fill it's octet. This will be modified throughout the site as the program will be run multiple times where the nonmetal will change values.** ● **non_metal_two** ○ **This represents the second nonmetal being bonded. Its type will be a dataclass Nonmetal which will include its Symbol and Valance Electrons. This will be modified throughout the site as the program will be run multiple times where the nonmetal will change values.** ● **valence_electrons** ○ **This represents the total number of electrons of the bonded compound. Its type will be an integer. This will be modified throughout the site depending on the bonded molecule and will be used to generate the lewis structure.**

	● bonded **○ This represents the chemical formula of the molecule. Its type will be a str. This will be modified throughout the program as different molecules will be modeled and this will be used to generate the 3D shape.** **● history** **○ This represents the molecules you have already created. Its type will be a list of a dataclass which will include fields of the bonded molecules like its name and image. This will be modified throughout the program as different molecules will be appended once they have been created.**
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	**The second dataclass will be defined as Nonmetal and its fields include:** **● symbol** **○ This will be a string representing the atomic symbol of the element. This will be modified as each element has a different atomic symbol.** **● valence_electrons** **○ This will be an integer representing the number of valence electrons the element has. This will be modified as each element has a different number of valence electrons.**
Routes (minimum 7) The names of your route functions	**1. determine_valid_nonmetal** **2. bond_with_flourine** **3. bond_with_oxygen** **4. bond_with_bromine** **5. bond_with_chlorine** **6. molecule_to_lewis** **7. lewis_to_structure** **8. new_compound** **9. show_history**

Pages (minimum 5) A brief description of the pages in your site	1. **no_valid_nonmetal_page** a. **Page when the user inputs an invalid nonmetal** 2. **select_nonmetal_page** a. **Page when the user inputs a valid nonmetal and has the option to choose what to bond the first nonmetal with** 3. **bonded_element_page** a. **Page when the element is bonded and the compound is listed. User will have the option to show its lewis structure** 4. **lewis_structure_page** a. **Page when the lewis structure of the molecule is shown. User will have the option to show its structure/3D model** 5. **three_d_page** a. **Page when the 3D model/structure is shown. User will have the option to create a new molecule** 6. **show_history_page** a. **Page that shows the history of the molecules created. The user will have the option to select a check box if they want to see the images of the molecules.**
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	1. **TextBox where the user inputs a nonmetal** 2. **SelectBox where the user decides what element the nonmetal will bond with** 3. **CheckBox where the user wants to include images of the molecule history**
If Statements (minimum 3) Brief description of what will be	1. **An if statement that will be evaluated within my program will be depending on the molecule, a certain image will be**

evaluated in your if statements and how they will meaningfully change the logic of the site	**inputted. Like if the molecule equals X then input this image.** **2. An if statement that will be evaluated within my program will be whether another molecule needs to be added depending on the total valence electrons.** **3. An if statement that will be evaluated within my program will be if this element is selected the molecule will be bonded in this specific molecule will be created.**
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	**1. A loop that will be iterated over this program is the show history which is a list of a dataclass. This is meaningful to the site as it will show the previous molecules made and shows what they look like.**

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM during this part of the project.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.